

Calc I Lecture 7: Derivatives and rates of change

Now we are at a position to revisit the concepts of instantaneous rate of change and tangent lines.

we have two perspectives: rates of change and slopes.

Perspective 1 (rates): Given a function $f(x)$,

- its average rate of change over the interval $[a, b]$ is $\frac{f(b) - f(a)}{b - a} \left\{ \begin{array}{l} \Delta f \\ \Delta x \end{array} \right.$
($\Delta = \text{"change"}$)

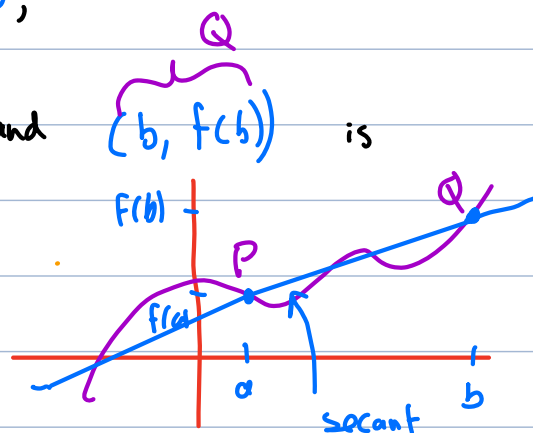
- its instantaneous rate of change at a is
let x represent b , and $x \rightarrow a$:

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Perspective 2 (slopes): Given a function $f(x)$,

- the slope of the secant line between $\overbrace{(a, f(a))}^P$ and $\overbrace{(b, f(b))}^Q$ is

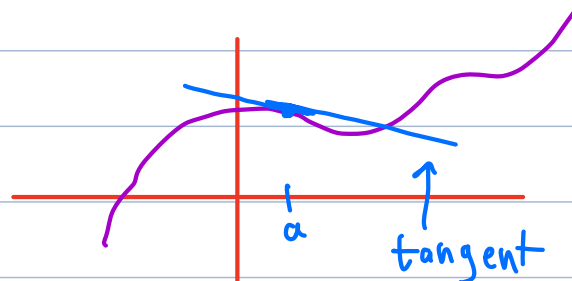
$$m_{PQ} = \frac{f(b) - f(a)}{b - a} \left\{ \begin{array}{l} \text{rise} \\ \text{run} \end{array} \right.$$



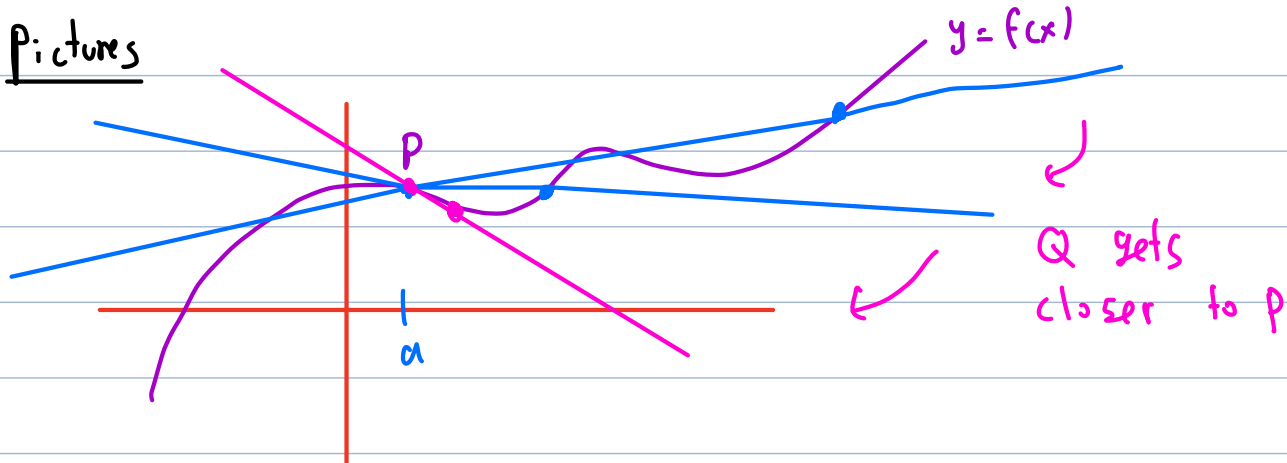
- the slope of the tangent line at a is

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

"limit of secant lines as $b \rightarrow a$ is tangent line"



Pictures



This limit that appears in both perspectives is what we call the derivative of the function f at a

Definition: The derivative of a function f at a point a

is
$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

(if this limit exists)

Equivalent expression for the limit:

If in the above expression we let $h = x - a$, then we get

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$\begin{aligned} &\underline{x = h + a} \\ &\text{as } x \rightarrow a \\ &h = x - a \rightarrow a - a = 0 \end{aligned}$$

Example Use the limit definition of derivative to find:

a) $f'(1)$, where $f(x) = x^2$

b) $f'(4)$, where $f(x) = \sqrt{x}$

Solutions:

a) $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$

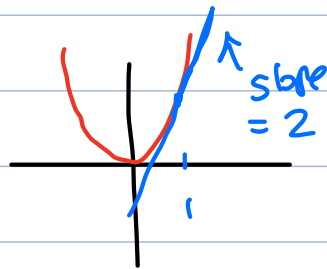
$f(x) = x^2, a = 1$

$$= \lim_{x \rightarrow 1} \frac{x^2 - 1^2}{x - 1}$$
$$= \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{x-1}$$
$$= \lim_{x \rightarrow 1} x+1 = 2$$

direct subst.

interpretations:

- tangent to $y = x^2$ at $x=1$ has slope 2
- rate of change of x^2 when $x=1$ is 2



b) $f'(4)$, where $f(x) = \sqrt{x}$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow 4} \frac{f(x) - f(4)}{x - 4}$$

$f(x) = \sqrt{x}$

$$a = 4$$

$$= \lim_{x \rightarrow 4} \frac{\sqrt{x} - \sqrt{4}}{x - 4} = \lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4}$$

multiply and divide
by conjugate

$$= \lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4} \cdot \frac{\sqrt{x} + 2}{\sqrt{x} + 2}$$

$$= \lim_{x \rightarrow 4} \frac{(\sqrt{x})^2 - 2^2}{(x - 4)(\sqrt{x} + 2)} = \lim_{x \rightarrow 4} \frac{\cancel{x} - 4}{(\cancel{x} - 4)(\sqrt{x} + 2)}$$

$$= \lim_{x \rightarrow 4} \frac{1}{\sqrt{x} + 2} = \frac{1}{\sqrt{4} + 2} = \frac{1}{2 + 2} = \boxed{\frac{1}{4}}$$

direct
Subst.
 $x = 4$

Definition The tangent line to the curve $y=f(x)$ at the point $P = (a, f(a))$ is the line through P with slope

$$m = f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

(Provided this limit exists)

Example Find an equation for the tangent line to the parabola $y = x^2$ at the point $P = (1, 1)$

use point-slope formula! point (x_0, y_0) slope $m \Rightarrow \boxed{y = y_0 + m(x - x_0)}$

Point: $(1, 1) = (x_0, y_0)$

slope: $f'(1) = 2$ (Previous problem part a)

$$y = 1 + 2(x - 1) = 1 + 2x - 2$$

$$\boxed{y = 2x - 1}$$

Idea: The tangent line to $f(x)$ at a approximates the behavior of the function near a

Speed and Velocity

Suppose $f(t)$ describes the position of an object at time t .

Then $f'(t)$ is velocity at time $t = v(t)$

$|f'(t)|$ is speed at time t

We can think of negative velocity as downward direction if travelling vertically
left direction if travelling horizontally

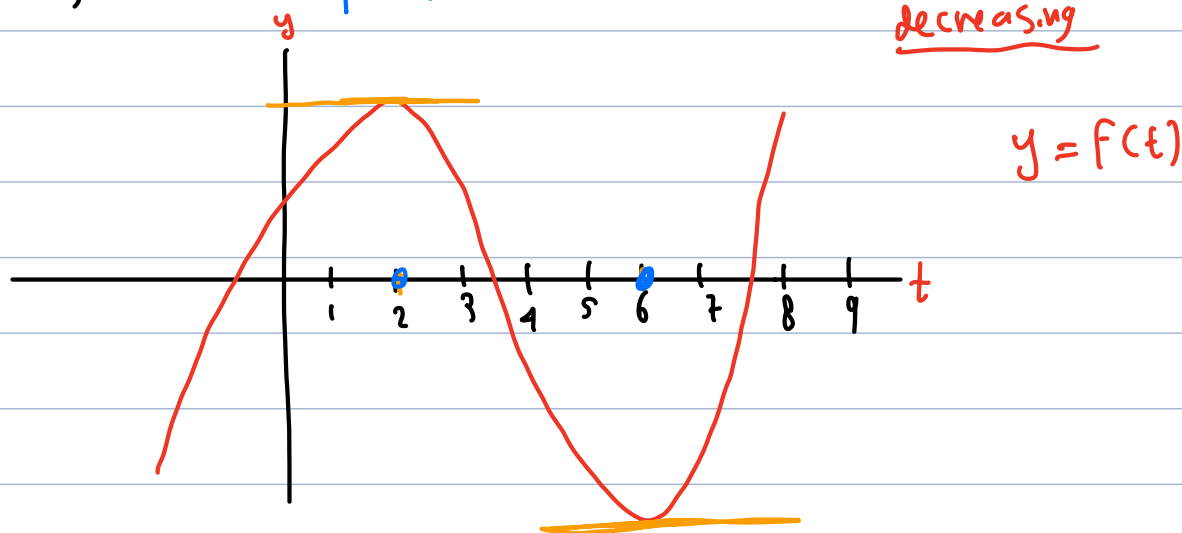
Positive velocity as up if vertically
right if horizontally

Example A graph of $f(t)$ is provided. Based on this graph,

a) When is $f'(t) = 0$? idea: $f' = 0 \Rightarrow$ slope 0 = horizontal tangent $\Rightarrow$ often local max or local min

b) When is $f'(t) > 0$? $f' > 0 \Rightarrow f$ is increasing (at the point of change between increasing / decreasing)

c) When is $f'(t) < 0$? $f' < 0 \Rightarrow f$ is decreasing

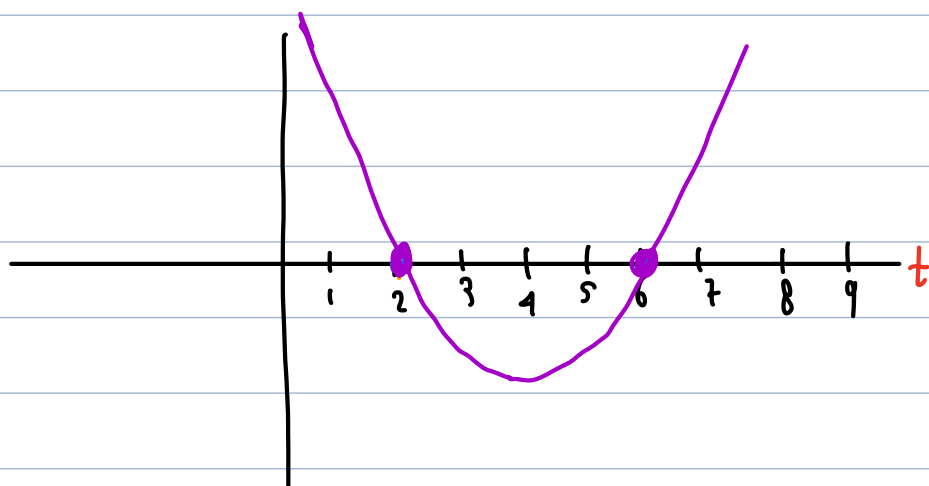


a) $t=2$ or $t=6$

b) $(-\infty, 2) \cup (6, \infty)$

c) $(2, 6)$

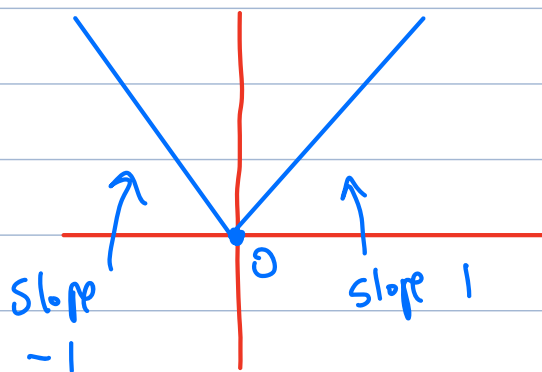
We can use this info to make a sketch of p' :



$$y = f'(t)$$

Example where the derivative is not defined

$$y = |x| = f(x)$$



no tangent line at 0

using limit def;

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} &= \lim_{x \rightarrow 0} \frac{|x| - |0|}{x - 0} \\ &= \lim_{x \rightarrow 0} \left(\frac{|x|}{x} \right) \text{ DNE} \end{aligned}$$

$$\frac{|x|}{x} = \begin{cases} \frac{x}{x} = 1 & x > 0 \\ -\frac{x}{x} = -1 & x < 0 \end{cases}$$

Recognizing limits as derivatives

Example: write the following limits as derivatives

a) $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = f'(a) \text{ for some } f(x) \text{ some } a$

b) $\lim_{x \rightarrow 0} \frac{\cos(\pi+x) + 1}{x}$

c) $\lim_{x \rightarrow 1} \frac{\frac{3}{x} - 3}{x - 1}$

Solutions

$$a) \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$$

$$= \lim_{h \rightarrow 0} \frac{\ln(1+h)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$\ln(1+h) = f(a+h) \Rightarrow \text{guess: } f(x) = \ln(x)$$

$$a = 1$$

check:

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{\ln(1+h) - \ln(1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\ln(1+h)}{h} \checkmark$$

$$c) \lim_{x \rightarrow 1} \frac{\frac{3}{x} - 3}{x - 1} \stackrel{?}{=} \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$\text{guess: } f(x) = \frac{3}{x}, a = 1.$$

$$\text{check: } \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow 1} \frac{\frac{3}{x} - \frac{3}{1}}{x - 1}$$

$$= \lim_{x \rightarrow 1} \frac{\frac{3}{x} - 3}{x - 1} \checkmark$$

Calculating derivatives from the definition

Recall we had two expressions for the derivative:

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

It turns out the second one is often (but not always) easier to use.

Example use the definition of the derivative to calculate the derivative $f'(a)$ at a general point a for the given functions:

a) $f(x) = x^2$

b) $f(x) = x^3 + x$

c) $f(x) = \frac{1}{x^2 - 1}$

d) $f(x) = \sqrt{x^2 - 1}$

a) $f(x) = x^2$

$$f'(a) = \lim_{h \rightarrow 0} \frac{\overbrace{f(a+h)}^{(a+h)^2} - \overbrace{f(a)}^{a^2}}{h} = \lim_{h \rightarrow 0} \frac{(a+h)^2 - a^2}{h}$$

← expand out

$$= \lim_{h \rightarrow 0} \frac{(\cancel{a^2} + 2ah + h^2) - \cancel{a^2}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2ah + h^2}{h} = \lim_{h \rightarrow 0} \frac{\cancel{h}(2a + h)}{\cancel{h}}$$

$$= \lim_{h \rightarrow 0} 2a + h = 2a + 0 = \boxed{2a}$$

direct
Subst.
 $h=0$

Conclusion: $\boxed{f'(a) = 2a}$

$$\boxed{f'(1) = 2}$$

b) $f(x) = x^3 + x$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$f(a+h) = (a+h)^3 + (a+h)$$

$$f(a) = a^3 + a$$

$$\rightarrow = \lim_{h \rightarrow 0} \frac{(\underline{(a+h)^3} + (a+h)) - (a^3 + a)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(\cancel{a^3} + \underline{3a^2h + 3ah^2 + h^3}) + \cancel{(a+h)} - (\cancel{a^3} + \cancel{a})}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3a^2h + 3ah^2 + h^3 + h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{h}(3a^2 + 3ah + h^2 + 1)}{\cancel{h}}$$

$$= \lim_{h \rightarrow 0} (3a^2 + 3ah + h^2 + 1) = \begin{matrix} \text{direct} \\ \text{subst.} \\ h=0 \end{matrix} \quad \begin{matrix} 3a^2 + 3a \cdot 0 + 0^2 + 1 \\ = \boxed{3a^2 + 1} \end{matrix}$$

$$c) f(x) = \frac{1}{x^2 - 1}$$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{(a+h)^2 - 1} - \frac{1}{a^2 - 1}}{h} \quad \left. \vphantom{\lim_{h \rightarrow 0}} \right\} \begin{matrix} \text{put over} \\ \text{common} \\ \text{denom.} \end{matrix}$$

$$= \lim_{h \rightarrow 0} \left[\frac{1}{(a+h)^2 - 1} \cdot \frac{a^2 - 1}{a^2 - 1} - \frac{1}{a^2 - 1} \cdot \frac{(a+h)^2 - 1}{(a+h)^2 - 1} \right] / h$$

$$= \lim_{h \rightarrow 0} \frac{(a^2 - 1) - ((a+h)^2 - 1)}{((a+h)^2 - 1)(a^2 - 1)} \cdot \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{a^2} - 1 - (\cancel{a^2} + 2ah + \cancel{h^2} - 1)}{((a+h)^2 - 1)(a^2 - 1)} \cdot \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-\cancel{h}(2a+h)}{(a+h)^2-1)(a^2-1)} \cdot \frac{\cancel{1}}{\cancel{h}}$$

$$= \lim_{h \rightarrow 0} \frac{-(2a+h)}{((a+h)^2-1)(a^2-1)}$$

$$\begin{aligned} &= \text{direct subst. } h=0 \\ &\frac{-(2a+0)}{(\underbrace{(a+0)^2-1}_{a^2-1})(a^2-1)} \end{aligned}$$

$$= \boxed{\frac{-2a}{(a^2-1)^2}} = f'(a)$$

$$d) f(x) = \sqrt{x^2-1}$$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{(a+h)^2-1} - \sqrt{a^2-1}}{h}$$

multiply top and bottom by conjugate

$$= \lim_{h \rightarrow 0} \frac{(\sqrt{(a+h)^2-1} - \sqrt{a^2-1})}{h} \cdot \frac{(\sqrt{(a+h)^2-1} + \sqrt{a^2-1})}{\sqrt{(a+h)^2-1} + \sqrt{a^2-1}}$$

$$= \lim_{h \rightarrow 0} \frac{\left(\sqrt{(a+h)^2-1}\right)^2 - \left(\sqrt{a^2-1}\right)^2}{h \left(\sqrt{(a+h)^2-1} + \sqrt{a^2-1}\right)}$$

$$= \lim_{h \rightarrow 0} \frac{((a+h)^2-1) - (a^2-1)}{h \left(\sqrt{(a+h)^2-1} + \sqrt{a^2-1}\right)}$$

$$= \lim_{h \rightarrow 0} \frac{(\cancel{a^2} + 2ah + \cancel{h^2} - 1) - (\cancel{a^2} - 1)}{h \left(\sqrt{(a+h)^2-1} + \sqrt{a^2-1}\right)}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{h}(2a+h)}{\cancel{h} \left(\sqrt{(a+h)^2-1} + \sqrt{a^2-1}\right)}$$

$$= \lim_{h \rightarrow 0} \frac{2a+h}{\sqrt{(a+h)^2-1} + \sqrt{a^2-1}} \quad \begin{array}{l} \text{direct} \\ \text{subst.} \\ h=0 \end{array} \quad \begin{array}{l} \frac{2a+0}{\underbrace{\sqrt{(a+0)^2-1} + \sqrt{a^2-1}}_{a^2-1}} \end{array}$$

$$= \frac{2a}{2\sqrt{a^2-1}} = \boxed{\frac{a}{\sqrt{a^2-1}}}$$

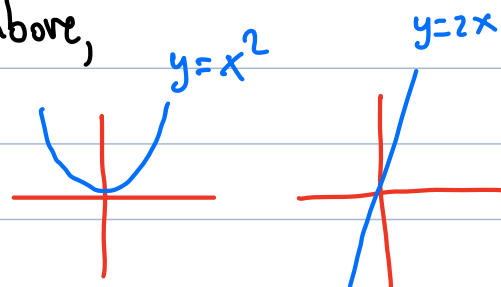
A change in perspective: the derivative as a function

If $f(x)$ is a function, then we get a new function $f'(x)$ derived from it, defined by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Example: As we see from the calculations above,

$$f(x) = x^2 \Rightarrow f'(x) = 2x$$



$$f(x) = x^3 + x \Rightarrow f'(x) = 3x^2 + 1$$

$$f(x) = \frac{1}{x^2 - 1} \Rightarrow f'(x) = \frac{-2x}{(x^2 - 1)^2}$$

$$f(x) = \sqrt{x^2 - 1} \Rightarrow f'(x) = \frac{x}{\sqrt{x^2 - 1}}$$

Another notation for derivatives

$$y = f(x) \Rightarrow$$

$$f'(x) = \frac{df}{dx} \quad \left(\approx \frac{\Delta f}{\Delta x} \right)$$

$$= \frac{dy}{dx}$$

e.g. $\frac{d}{dx}(x^3 + x) =$ derivative of $x^3 + x$

Higher derivatives

Given $f(x)$, we get $f'(x) = \frac{df}{dx}$

$$f''(x) = \frac{d}{dx} (f'(x))$$

$$f'''(x) = \frac{d}{dx} (f''(x))$$

$\vdots$

$$f^{(n)}(x) = n\text{th derivative}$$

$$f^{(10)}(x) = 10\text{th derivative}$$

Application: if $f(t) = \text{position in time} = s(t)$

$$f'(t) = \text{velocity} = v(t)$$

$$f''(t) = \text{acceleration} = a(t)$$

$$f'''(t) = \text{jerk}$$

Differentiability:

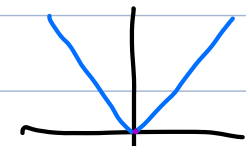
$f(x)$ is differentiable at a if

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \text{ exists}$$

What might make a function not differentiable

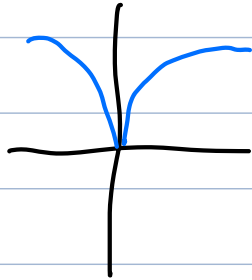
- Corner or cusp

e.g. $y = |x|$

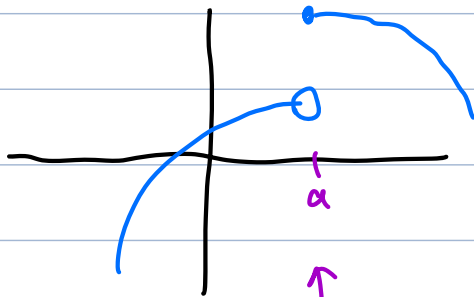


not differentiable at 0

e.g. $y = x^{2/3}$



- if $f(x)$ is not continuous at a , then it is not differentiable at a

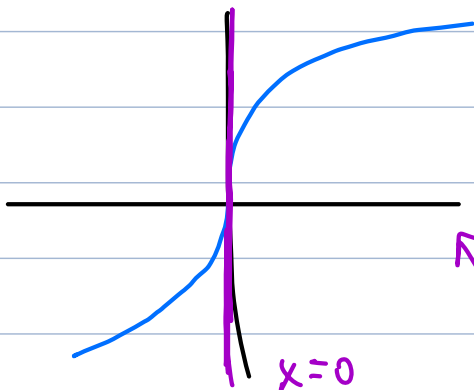


not differentiable at a

Caution: differentiable $\Rightarrow$ continuous
continuous does not imply differentiable

- Vertical tangents:

"infinite slope" $\Rightarrow$ slope undefined

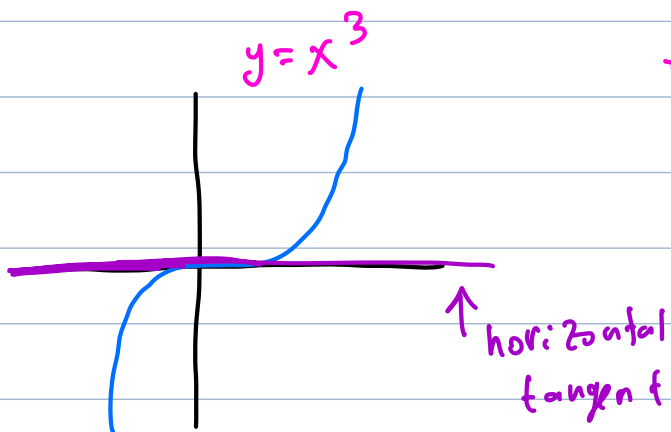


vertical tangent

more precisely: f has a vertical tangent if f' has a vertical asymptote

Example:

$$y = x^{1/3}$$



take
inverse
→

$$y = x^{1/3} = \sqrt[3]{x}$$

